

Performance Testing

ComicCraft – AI Comic Story Creator using Gemini Models

Performance testing is used to verify that ComicCraft responds efficiently during the main comic-generation workflow. The testing focuses on request handling, AI generation, panel processing, preview loading and PDF export.

The test plan covers the complete workflow from opening the application and submitting a story prompt through AI-based content generation, panel creation, comic preview and final PDF export. It also considers repeated valid requests and situations in which an AI request cannot be completed.

Performance Test Cases

S.No	Test Area	Performance Check	Expected Result
1	Application Loading	Open the ComicCraft application and load the main page. Observe whether the application becomes available without unnecessary delay.	The application loads successfully without unnecessary delay and is ready for user interaction.
2	Story Generation	Submit a valid story prompt with character, setting, tone and art style. Observe the request-processing behaviour and returned story structure.	The request is processed and the generated story structure is returned successfully.
3	AI Response Handling	Observe response handling during Gemini-based story generation, including the normal successful response path and a generation failure condition.	The application handles successful responses and reports errors clearly when generation fails.
4	Panel Generation	Generate illustrations for the required comic panels and observe whether panel processing completes correctly without unnecessarily blocking the overall workflow.	Panel images are generated and stored correctly without blocking the overall workflow unnecessarily.
5	Comic Preview	Open the generated comic preview and inspect the rendering of panels, narration and dialogue.	Panels, narration and dialogue are displayed correctly with smooth page rendering.
6	PDF Export	Export the completed comic as a PDF and verify that the generated document contains the expected comic content.	The PDF is generated successfully and contains the expected comic content.
7	Multiple Requests	Repeat the comic-generation workflow using different valid inputs and observe whether the application continues to respond correctly.	The application continues to respond correctly without functional degradation.
8	Error / Quota Handling	Test the workflow when an AI request cannot be completed and observe how the application handles the failed request.	A clear error message is shown and the application does not crash.

Performance Testing Areas in Detail

Application Responsiveness

The initial application-loading check verifies that the main ComicCraft page becomes available without unnecessary delay. This establishes the starting point for the rest of the user workflow.

AI Generation Handling

Story generation is checked using a valid prompt containing character, setting, tone and art style. The focus is on successful request processing and clear handling when Gemini-based generation cannot be completed.

Panel Processing

Panel generation is checked to ensure that required illustrations are produced and stored correctly. The test also considers whether panel processing unnecessarily blocks the overall workflow.

Preview Rendering

The generated comic preview is checked for correct display of panels, narration and dialogue. Smooth page rendering is the expected behaviour.

Export Processing

The completed comic is exported as a PDF. The generated document is checked for successful creation and inclusion of the expected comic content.

Repeated Workflow

The complete generation workflow is repeated with different valid inputs. The purpose is to verify that the application continues to respond correctly without functional degradation.

Error and Quota Conditions

The test plan includes a condition in which an AI request cannot be completed. The expected behaviour is a clear error message without an application crash.

Overall Workflow Coverage

The performance test plan covers the complete ComicCraft workflow: application loading, story generation, AI response handling, panel generation, comic preview, PDF export, repeated requests, and error/quota handling. Together, these checks provide coverage of both normal and error conditions across the main user-facing workflow.

Performance Testing Summary

The performance test plan covers the complete ComicCraft workflow from application loading through AI story generation, panel creation, preview and PDF export. The objective is to ensure that the application remains responsive and handles normal and error conditions in a controlled manner.

Conclusion

Performance testing helps validate the responsiveness and stability of ComicCraft during its major user workflows. The identified checks provide a structured basis for verifying application behaviour before final delivery.